Assignment 2

(Deadline: Nov. 10. **Late submission will be considered as 0 point.**)

1. **(30 points)**
(1) Construct a k-d tree with the following nodes with a root (30; 40):
(30; 40); (5; 25); (10; 12); (70; 70); (50; 30); (35; 45)
(2) Construct a k-d tree with the following nodes with a root (8; 6):
(8; 6); (10; 1); (5; 8); (9; 7); (2; 1); (3; 5); (1; 7); (7; 10); (2; 9); (6; 2)

2. **(30 points)** Calculate the first 10 elements of the Halton sequence in 3D. For simplicity, you can use the smallest 3 prime numbers as the base.

3. **(40 points)**
(1) **(15 points)** Recall the concept of admissible heuristic. If h_1 and h_2 are both admissible, which of the following heuristics are also guaranteed to be admissible?
(a) $h_1 + h_2$;
(b) $h_1 * h_2$;
(c) $\max(h_1, h_2)$;
(d) $\min(h_1, h_2)$;
(e) $\alpha h_1 + (1-\alpha)h_2, \alpha \in [0,1]$.
(2) **(25 points)** (*True* or *False.* A concise explanation is required.) Consider an optimal path that is calculated with Uniform-Cost Search. If we add a positive constant to every step cost, the optimal path may change.